

Autodesk® Scaleform®

Scale9Grid

Scaleform

Scale9Grid

Maxim Shemanarev, Michael Antonov

1.01

2008 3 21

Copyright Notice

Autodesk® Scaleform® 4.4

© 2014 Autodesk, Inc. All rights reserved. Except as otherwise permitted by Autodesk, Inc., this publication, or parts thereof, may not be reproduced in any form, by any method, for any purpose.

Certain materials included in this publication are reprinted with the permission of the copyright holder.

The following are registered trademarks or trademarks of Autodesk, Inc., and/or its subsidiaries and/or affiliates in the USA and other countries: 123D, 3ds Max, Algor, Alias, AliasStudio, ATC, AutoCAD LT, AutoCAD, Autodesk, the Autodesk logo, Autodesk 123D, Autodesk CAM 360, Autodesk Homestyler, Autodesk Inventor, Autodesk MapGuide, Autodesk Streamline, AutoLISP, AutoSketch, AutoSnap, AutoTrack, Backburner, Backdraft, Beast, BIM 360, Burn, Buzzsaw, CADmep, CAiCE, CAMduct, CFdesign, Civil 3D, Cleaner, Combustion, Communication Specification, Configurator 360™, Constructware, Content Explorer, Creative Bridge, Dancing Baby (image), DesignCenter, DesignKids, DesignStudio, Discreet, DWF, DWG, DWG (design/logo), DWG Extreme, DWG TrueConvert, DWG TrueView, DWGX, DXF, Ecotect, ESTmep, Evolver, FABmep, Face Robot, FBX, Fempro, Fire, Flame, Flare, Flint, FMDesktop, ForceEffect, FormIt, Freewheel, Fusion 360, Glue, Green Building Studio, Heidi, Homestyler, HumanIK, i-drop, ImageModeler, Incinerator, Inferno, InfraWorks, InfraWorks 360, Instructables, Instructables (stylized robot design/logo), Inventor, Inventor HSM, Inventor LT, Kynapse, Kynogon, LandXplorer, Lustre, MatchMover, Maya, Maya LT, Mechanical Desktop, MIMI, Mockup 360, Moldflow Plastics Advisers, Moldflow Plastics Insight, Moldflow, Moondust, MotionBuilder, Movimento, MPA (design/logo), MPA, MPI (design/logo), MPX (design/logo), MPX, Mudbox, Navisworks, ObjectARX, ObjectDBX, Opticore, Pipeplus, Pixlr, Pixlr-o-matic, Productstream, Publisher 360, RasterDWG, RealDWG, ReCap, ReCap 360, Remote, Revit LT, Revit, RiverCAD, Robot, Scaleform, Showcase, Showcase 360 ShowMotion, Sim 360, SketchBook, Smoke, Socialcam, Softimage, Sparks, SteeringWheels, Stitcher, Stone, StormNET, TinkerBox, ToolClip, Topobase, Toxik, TrustedDWG, T-Splines, ViewCube, Visual LISP, Visual, VRED, Wire, Wiretap, WiretapCentral, XSI.

All other brand names, product names or trademarks belong to their respective holders.

Disclaimer

THIS PUBLICATION AND THE INFORMATION CONTAINED HEREIN IS MADE AVAILABLE BY AUTODESK, INC. "AS IS." AUTODESK, INC. DISCLAIMS ALL WARRANTIES, EITHER EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE REGARDING THESE MATERIALS.

Autodesk Scaleform

Scale9Grid User Guide

Autodesk Scaleform Corporation
6305 Ivy Lane, Suite 310
Greenbelt, MD 20770, USA

www.scaleform.com

info@scaleform.com

(301) 446-3200

(301) 446-3199

1	Flash	Scaleform	Scale9Grid.....	1
2				2
3				2
4				5
4.1	Flash			5
4.2		Scaleform		9

1 Flash Scaleform Scale9Grid

Scale9Grid Flash® Scale9Grid

Flash Scale9Grid 9-Slice Scaling

Flash Studio "Symbol properties" 9-Slice Scaling
 ActionScript MovieClip.scale9Grid Button.scale9Grid
 Flash Scale9Grid Flash Studio

Adobe Flash Scale9Grid
 Flash Scale9Grid Scaleform
 Scale9Grid Adobe Flash
 Scaleform

Scale9Grid	Adobe Flash	Scaleform
	X/Y	
		Scale9Grid
Scale9Grid		

Scaleform Flash Flash Scale9Grid Scaleform
 Flash Scale9Grid

Scaleform Scale9Grid FLA/SWF

2

Flash Scaleform Scale9Grid Flash

 Sprite

 Flash Scaleform

 Scale9Grid

3

 Flash Scaleform

 Flash

Scale9Grid Scaleform

 Scale9Grid

 EdgeAA

Scale9Grid
Scale9Grid

Flash
Scale9Grid

Scaleform

Adobe Flash

4

ActionScript

4.1 *Flash*

Flash
scale0grid_window1 fla scale9grid_window1 swf Flash Studio

Flash ActionScript hit-testing Flash

ActionScript

```
import flash.geom.Rectangle;

var bounds:Object = this.getBounds(this);
for (var i in bounds) trace(i+" --> "+bounds[i]);
this.XMin = bounds.xMin;
this.YMin = bounds.yMin;
this.XMax = bounds.xMax;
this.YMax = bounds.yMax;
this.MinW = 120;
this.MinH = 100;
this.OldX = this._x;
this.OldY = this._y;
this.OldW = this._width;
this.OldH = this._height;
this.OldMouseX = 0;
this.OldMouseY = 0;
this.XMode = 0;
this.YMode = 0;
```

```

this.onPress = function()
{
    this.OldX = this._x;
    this.OldY = this._y;
    this.OldW = this._width;
    this.OldH = this._height;
    this.OldMouseX = _root._xmouse;
    this.OldMouseY = _root._ymouse;
    this.XMode = 0;
    this.YMode = 0;

    var kx = (this.XMax - this.XMin) / this._width;
    var ky = (this.YMax - this.YMin) / this._height;
    var x1 = this.XMin + 6 * kx;    // Left
    var y1 = this.YMin + 4 * ky;    // Top
    var x2 = this.XMax - 26 * kx;   // Right
    var y2 = this.YMax - 25 * ky;   // Bottom
    var w = 20 * kx;
    var h = 20 * ky;
    var xms = this._xmouse;
    var yms = this._ymouse;

    if (xms >= x1 && xms <= x1+w)
    {
        if (yms >= y1 && yms <= y1+h)
        {
            this.XMode = -1;
            this.YMode = -1;
        }
        else
        if (yms >= y2 && yms <= y2+h)
        {
            this.XMode = -1;
            this.YMode = 1;
        }
    }
    else
    if (xms >= x2 && xms <= x2+w)
    {
        if (yms >= y1 && yms <= y1+h)
        {
            this.XMode = 1;
            this.YMode = -1;
        }
        else
        if (yms >= y2 && yms <= y2+h)
        {
            this.XMode = 1;
            this.YMode = 1;
        }
    }

    if (XMode == 0 && YMode == 0)
    {
        this.startDrag();
    }
}

```

```

}

this.onRelease = function()
{
    this.XMode = 0;
    this.YMode = 0;
    this.stopDrag();
}

this.onReleaseOutside = function()
{
    this.XMode = 0;
    this.YMode = 0;
    this.stopDrag();
}

this.onMouseMove = function()
{
    var dx = _root._xmouse - OldMouseX;
    var dy = _root._ymouse - OldMouseY;

    if (this.XMode == -1)
    {
        this._x      = this.OldX + dx;
        this._width = this.OldW - dx;
        if (this._width < this.MinW || _root._xmouse > this.OldX + this.OldW)
        {
            this._x      = this.OldX + this.OldW - this.MinW;
            this._width = this.MinW;
        }
    }
    if (this.XMode == 1)
    {
        this._width = this.OldW + dx;
        if (this._width < this.MinW || _root._xmouse < this.OldX)
            this._width = this.MinW;
    }
    if (this.YMode == -1)
    {
        this._y      = this.OldY + dy;
        this._height = this.OldH - dy;
        if (this._height < this.MinH || _root._ymouse > this.OldY + this.OldH)
        {
            this._y      = this.OldY + this.OldH - this.MinH;
            this._height = this.MinH;
        }
    }
    if (this.YMode == 1)
    {
        this._height = this.OldH + dy;
        if (this._height < this.MinH || _root._ymouse < this.OldY)
            this._height = this.MinH;
    }
}

```

"1" XMode Ymode "-1"

onPress () kx ky
Scale9Grid

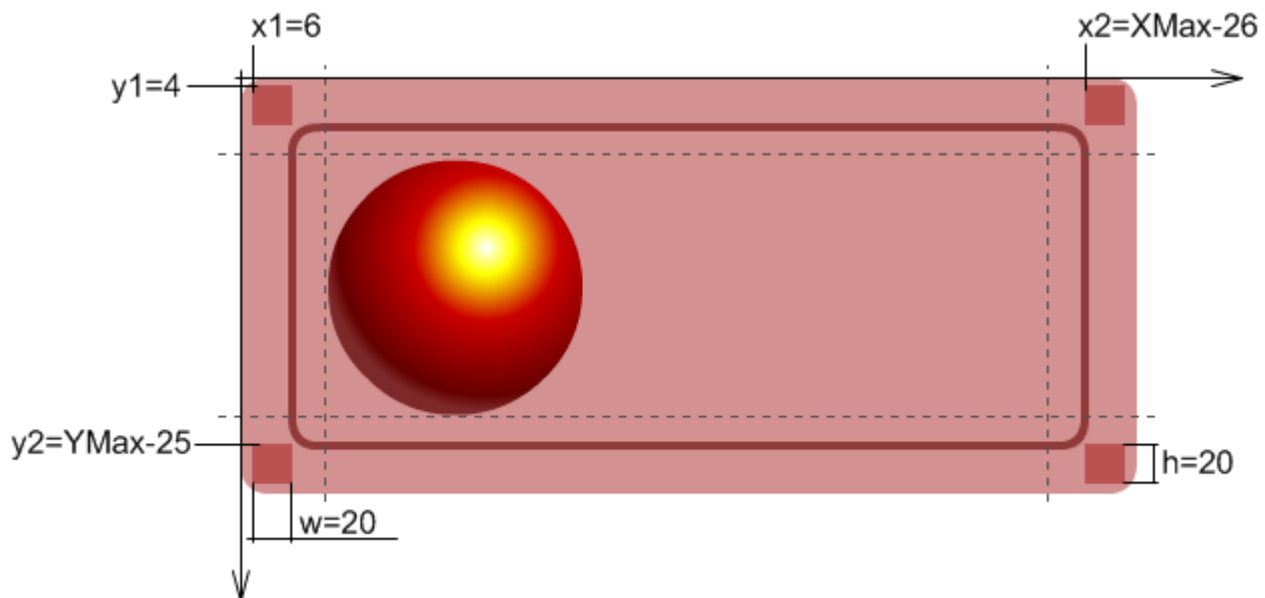
```
var kx = (this.XMax - this.XMin) / this._width;  
var ky = (this.YMax - this.YMin) / this._height;
```

Left-Top: x1, y1, x1+w, y1+h
Right-Top: x2, y1, x2+w, y1+h
Left-Bottom: x1, y2, x1+w, y2+h
Right-Bottom: x2, y2, x2+w, y2+h

```
var x1 = this.XMin + 6 * kx; // Left  
var y1 = this.YMin + 4 * ky; // Top  
var x2 = this.XMax - 26 * kx; // Right  
var y2 = this.YMax - 25 * ky; // Bottom  
var w = 20 * kx;  
var h = 20 * ky;
```

6 4 26 25

ActionScript



```
var xms = this._xmouse;
```

```

var yms = this._ymouse;
if (xms >= x1 && xms <= x1+w)
{
    ... and so on

```



ActionScript

Flash

4.2 **Scaleform**

Scaleform

Flash

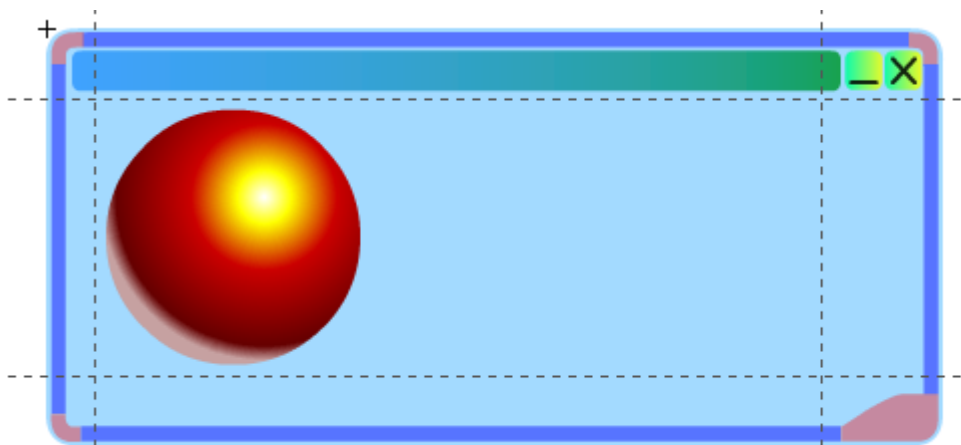
Scaleform

scale9grid_window2.fla

scale9grid_window2.swf

ActionScript

" "



- Border1
- Border2
- Border3
- Border4
- BtnClose
- BtnMinimize
- Corner1
- Corner2
- Corner3
- Corner4
- TitleBar
- Window

Flash			Border1	Border4	Corner1
Corner4	TitleBar	Scaleform	Scale9Grid		
		ActionScript			OnResize ()

```

import flash.geom.Rectangle;
//trace(this.scale9Grid);
this.MinW = 100;
this.MinH = 100;
this.XMode = 0;
this.YMode = 0;
this.OldX = this._x;
this.OldY = this._y;
this.OldW = this._width;
this.OldH = this._height;
this.OldMouseX = 0;
this.OldMouseY = 0;

function AssignResizeFunction(mc:MovieClip, xMode:Number, yMode:Number)
{
    mc.onPress          = function() { this._parent.StartResize(xMode, yMode); }
    mc.onRelease        = function() { this._parent.StopResize(); }
    mc.onReleaseOutside = function() { this._parent.StopResize(); }
    mc.onMouseMove      = function() { this._parent.OnResize(); }
}

AssignResizeFunction(this.Border1, 0, -1);
AssignResizeFunction(this.Border2, 1, 0);
AssignResizeFunction(this.Border3, 0, 1);
AssignResizeFunction(this.Border4, -1, 0);
AssignResizeFunction(this.Corn1, -1, -1);
AssignResizeFunction(this.Corn2, 1, -1);
AssignResizeFunction(this.Corn3, 1, 1);
AssignResizeFunction(this.Corn4, -1, 1);
this.TitleBar.onPress          = function() { this._parent.startDrag(); }
this.TitleBar.onRelease        = function() { this._parent.stopDrag(); }
this.TitleBar.onReleaseOutside = function() { this._parent.stopDrag(); }

function StartResize(xMode:Number, yMode:Number)
{
    this.XMode = xMode;
    this.YMode = yMode;
    this.OldX = this._x;
    this.OldY = this._y;
    this.OldW = this._width;
    this.OldH = this._height;
    this.OldMouseX = _root._xmouse;
    this.OldMouseY = _root._ymouse;
}

function StopResize()
{
    this.XMode = 0;
    this.YMode = 0;
}

```

```

function OnResize()
{
    var dx = _root._xmouse - OldMouseX;
    var dy = _root._ymouse - OldMouseY;

    if (this.XMode == -1)
    {
        this._x      = this.OldX + dx;
        this._width = this.OldW - dx;
        if (this._width < this.MinW || _root._xmouse > this.OldX + this.OldW)
        {
            this._x      = this.OldX + this.OldW - this.MinW;
            this._width = this.MinW;
        }
    }
    if (this.XMode == 1)
    {
        this._width = this.OldW + dx;
        if (this._width < this.MinW || _root._xmouse < this.OldX)
            this._width = this.MinW;
    }
    if (this.YMode == -1)
    {
        this._y      = this.OldY + dy;
        this._height = this.OldH - dy;
        if (this._height < this.MinH || _root._ymouse > this.OldY + this.OldH)
        {
            this._y      = this.OldY + this.OldH - this.MinH;
            this._height = this.MinH;
        }
    }
    if (this.YMode == 1)
    {
        this._height = this.OldH + dy;
        if (this._height < this.MinH || _root._ymouse < this.OldY)
            this._height = this.MinH;
    }
}
}

```

Adobe Flash

ActionScript

" "

	Flash Studio	Scale9Grid	SWF	BUG
scale9grid_window2 fla		BUG	"TitleBar"	
Scale9Grid	Flash Studio			
	Scale9Grid			"Save and Compact"
	ActionScript	Scale9Grid		

```
trace(this.scale9Grid);
```

BUG

```
this.scale9Grid = new Rectangle(24, 35, 363, 138);
```

scale9grid_window3 fla scale9grad_window3.swf

